2025 Ch H1 Q10

Section: Nature's Chemistry

Topic: Soaps, Detergents and Emulsions

Question summary

Saponification is the reaction between edible fats and sodium hydroxide to produce soaps and glycerol. Which type of reaction is this?

Worked solution

The reaction of a fat (an ester) with sodium hydroxide breaks the ester bonds.

This process splits the ester into an alcohol (glycerol) and the sodium salts of fatty acids (soaps).

Since the ester is broken down by reaction with water or hydroxide ions, this is a hydrolysis reaction.

Final answer

B. hydrolysis

Revision tips

- Esterification: acid + alcohol -> ester + water.

- Hydrolysis: ester + water (or alkali) -> acid + alcohol.

- In saponification, alkali hydrolysis of fats produces soap and glycerol — an important industrial process.